

Implement a face detection algorithm using Open CV to detect and locate human faces in the images.

Code:

```
import cv2, urllib.request

from google.colab.patches import cv2_imshow

# Download sample image
urllib.request.urlretrieve(
    "https://raw.githubusercontent.com/opencv/opencv/master/samples/data/lena.jpg",
    "face.jpg"
)

# Download Haar Cascade file
urllib.request.urlretrieve(
    "https://raw.githubusercontent.com/opencv/opencv/master/data/haarcascades/haarcascade_frontalface_default.xml",
    "face.xml"
)

# Read image
img = cv2.imread("face.jpg")
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Load classifier
face_cascade = cv2.CascadeClassifier("face.xml")

# Detect faces
faces = face_cascade.detectMultiScale(gray, 1.1, 4)

# Draw rectangle around faces
for (x, y, w, h) in faces:
    cv2.rectangle(img, (x, y), (x+w, y+h), (255,0,0), 2)

# Display result
print("Face Detection")
cv2_imshow(img)
```

Output:

Face Detection

